Lecture3

September 9, 2026

[15]:
```python
# Lecture 3

# Exercise of an alternating series
stuff = []
for i in range(1, 1015):
    thing = ((-1)**(i+1)*(1/(2*i-1)))
    stuff.append(thing)

doodoo = sum(stuff)
print(4*doodoo)
```

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3.140606460535695
```

[29]:
```python
# Nested for loops and nested lists

# List with a list in it, 2x3
A=[[0,0,0],[0,0,0]]

for i in range(len(A)):
    for j in range(len(A[0])):

        # [0 0 0]
        # [0 0 0]
        # ith entry and jth entry
        A[i][j] = i * j

print(A)
```

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[[0, 0, 0], [0, 1, 2]]
```

[39]:
```python
# One more example
a1 = 0
a_list = []

# get 100
for i in range(1, 100):

    a1 = a1 + i
```

```python
    a_list.append(a1)

print(a_list[-1])
```

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4950
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[43]:
```python
x = float(input('Enter the value x:'))
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Enter the value x: -5.6
```

[45]:
```python
print(x)
```

```
-5.6
```

[49]:
```python
if x >= 0:
    print(f"{x} is positive")
else:
    print(f"{x} is negative")
```

```
-5.6 is negative
```

[57]:
```python
# Practce
# Given park visitor's age: input()

age = int(input("Enter visitor's age"))

if age <= 3:
    fee = 0
elif age < 18:
    fee = 10
elif age < 65:
    fee = 15
else:
    fee = 5

print(f"Visitor's fee: {fee}")
```

```
Enter visitor's age 64
Visitor's fee: 15
```

[ ]:
```python

```